SUPER APP

Product Requirements Document (PRD)

Super App — the Egyptian sports super app: pitches, players, fantasy, shop, print on demand, and the systems that run the company.

Document	PRD — backend phase
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1. About this document

1.1 Purpose

Super App already exists as a complete front end (HTML, CSS, vanilla JavaScript) that stores its data in the browser. This document defines **what the backend must do**, so the same product runs on a real server with a real database, real accounts, real money flows and real integrations. It is the reference for engineers who build, test and review the backend.

The companion documents are:

Document	Answers
PRD (this one)	What the product does, for whom, and the rules it must follow
ERD	Every table, column, relationship, constraint and index
API	Every endpoint, who may call it, request and response shapes
INTEGRATIONS	Payments, couriers, email, SMS, football data, n8n, AI agents, e-receipts
STRUCTURE	Solution layout, packages, how to run, migrate, seed and test

1.2 Decisions already made (fixed rules)

Area	Decision
Backend	**ASP.NET Core Web API on .NET 10** (latest LTS), object-oriented, clean code, dependency injection everywhere
Pattern	Controllers → Services → Repositories, with Middlewares for authentication, authorization and validation, and one **global error handler**
Database	**Microsoft SQL Server**, **EF Core Code-First**. The owner runs `Add-Migration` and `Update-Database` himself
Front end	The existing site is served unchanged from `wwwroot`; its `js/services/*` files are switched from browser storage to API calls one by one. **The design must not change.**
Sign-in	**ASP.NET Core Identity** with a **secure cookie** (HttpOnly, Secure, SameSite). Two-factor sign-in is mandatory for the CEO and admins
First user	A default **CEO** account is created by the seed
Approvals	The CEO approves pitch owners, sellers and factories. The CEO can grant the **Approvals** permission to chosen admins
Admins	Created **directly by the CEO** with their permissions. Admins never sign up as customers
Seed data	Today's demo data becomes the seed, so the site is not empty on day one. The owner then adds real pitches and products
AI	AI agents **really run** (Claude + n8n). The backend gives them scoped access, events, approvals and full logs

Area	Decision
Football data	**API-FOOTBALL, free plan** first. No club logos for now. A usage bar warns before the free quota runs out
Chat	Support chat (customer ↔ Super App team) is live. **No private messages between customers** — they talk in communities
Email	One official company address sends everything. Until the domain exists, a placeholder address is used and, in development, emails are saved as files
Hosting	Local (SQL Server LocalDB) → free .NET hosting for testing → **Hostinger VPS** for launch
Not yet	Commercial register, tax card and ETA e-receipts: prepared in code but switched off

1.3 How to read the requirements

Each requirement has an ID such as **FR-BOOK-04** (functional) or **NFR-SEC-02** (non-functional). "Must" is mandatory for launch; "Should" is expected unless agreed otherwise; "Later" is designed for but built after launch.

2. Product overview

2.1 Vision

One app for everything football in Egypt: book a pitch, fill your squad, play fantasy, buy gear, and sell your own designs — run by a small human team with AI agents doing most of the daily work.

2.2 The seven systems

#	System	Path	Who uses it	Purpose
1	**Public site**	`/`	Customers (players, fans, shoppers)	Pitches, bookings, find players, tournaments, matches and news, fantasy, shop, communities, help
2	**Partner dashboard**	`/dash`	Pitch owners and their staff	Pitches, availability, bookings, QR check-in, packages, offers and codes, tournaments, free slots, reviews, store, finance, wallet
3	**Seller dashboard**	`/seller`	Sellers (print on demand and/or own products)	Designs, studio, orders, store settings, codes, analytics, marketing calendar, wallet, returns
4	**Factory system**	`/factory`	Factory owner and floor staff	Print jobs, stages, quality check, packing, courier shipping, team, money, reports
5	**Admin console**	`/admin`	Super App team (human admins and AI agents)	Approvals, money, people, orders, support, content, marketing, fantasy, communities, catalogue, factories, analytics, audit
6	**CEO Command Center**	`/ceo`	The CEO (and the CFO for second approvals)	Approvals over limits, org and permissions, admins, AI agents, integrations and APIs, kill switches, goals, budgets, reports, audit
7	**AI & automation layer**	(API only)	AI agents (Claude), n8n workflows	Scoped service accounts, events, tasks, human approvals, logs, costs

The systems share **one database and one API**. A change in one system is visible in the others immediately (for example, a pitch approved in the admin console appears on the site, and a booking made on the site appears in the partner dashboard).

2.3 Scope

In scope for the backend phase

- Accounts, roles, permissions, applications and approvals for every system.
- Every feature that exists in the front end today, moved to the server with the same behaviour and the same business rules (section 6).
- Payments, payouts, courier and email behind adapters, so a provider is plugged in when its contract is signed.
- Real football data and real fantasy scoring from API-FOOTBALL.

- Live support chat and live notifications.
- The integrations registry, API catalogue and AI-agent panel in the CEO system.
- Security, performance, logging, backups, seed data, tests and a Postman collection.

Out of scope for now (designed for, built later)

- Native mobile apps (the site becomes an installable PWA after the backend).
- Freelancer services marketplace (after launch).
- ETA e-receipts (after the commercial register and tax card).
- Club logos (after licensing).
- SMS and WhatsApp sending (after a provider is chosen).

3. Users, roles and permissions

3.1 Account types

Everyone has **one account** (one email, one phone). What an account can do comes from its **roles** and **permissions**.

Role	How it is obtained	Signs in to
Customer	Signs up on the site (email or phone + password, or a one-time code later)	Site
Pitch owner	A customer applies at `/partner/apply`; approved by the CEO or an admin with the Approvals permission	Site + `/dash`
Pitch staff	Created by a pitch owner, limited to that owner's pitches	`/dash` (limited)
Seller	A customer (or pitch owner) applies at `/sell` or `/dash/sell`; approved as above	Site + `/seller`
Factory owner	Applies through a factory application (link shared by Super App); approved by the CEO	`/factory`
Factory staff	Created by the factory owner: supervisor, printer, quality check, packer. Sign in with name + 4-digit PIN on a factory device	`/factory`
Admin	**Created by the CEO** with a set of permissions; receives an invitation email to set a password and two-factor sign-in	`/admin`
CFO	An admin with the Finance permissions and the right to give the **second approval**	`/admin`, `/ceo` (approvals only)
CEO	Created by the seed. All permissions. Cannot be deleted or suspended by anyone else	Everything
AI agent	Created by the CEO in the AI panel; a service account with an API key, permissions and a money limit	API only

One person can hold several roles (for example a pitch owner who is also a seller). One button switches between their dashboards.

3.2 Permission model

- Permissions are **fine-grained codes** grouped by area, for example `approvals.partners`, `payouts.prepare`, `payouts.send`, `refunds.issue`, `content.banners`, `community.moderate`, `fantasy.manage`, `settings.platform`, `audit.read`.
- **Roles** are named bundles of permissions (for example "Admin manager", "Support", "Finance"). The CEO creates and edits roles and assigns them to admins and AI agents.
- **Scopes:** partner, seller and factory permissions are always limited to the records they own (their pitches, their designs, their jobs). This is enforced on the server for every request, not only hidden in the UI.

- **Money limits** are attached to a role or a person: refunds and credits up to X EGP, payouts up to Y EGP. Above the limit, the action becomes an **approval request**.
- **Separation of duties** (enforced): preparing a payout ≠ sending it; adjusting a balance ≠ approving it; creating promo codes ≠ reviewing revenue; nobody can edit or delete the audit log.

3.3 Requirements — accounts and access

ID	Requirement	Priority
FR-AUTH-01	Customers sign up with name, email, phone (Egyptian format) and password; email verification by link; phone verification by code when an SMS provider exists	Must
FR-AUTH-02	Sign-in issues a secure HttpOnly cookie; sessions expire after inactivity (14 days for customers, 8 hours for admins and the CEO); sign out everywhere is available	Must
FR-AUTH-03	Account lockout after 5 failed attempts for 15 minutes; rate limits on sign-in, sign-up and password reset	Must
FR-AUTH-04	Password reset by email link (single use, 30-minute expiry)	Must
FR-AUTH-05	Two-factor sign-in (authenticator app) is mandatory for the CEO and every admin	Must
FR-AUTH-06	The seed creates the CEO account; its password and email come from environment variables, never from code	Must
FR-AUTH-07	The CEO creates admin accounts (name, work email, role, permissions, money limits); the admin gets an invitation email to set a password	Must
FR-AUTH-08	The CEO (or an admin with the right permission) can suspend any non-CEO account; suspension signs the account out and blocks sign-in	Must
FR-AUTH-09	Factory staff sign in on a registered factory device with their name and a 4-digit PIN; PINs are hashed and rate-limited	Must
FR-AUTH-10	Customers can download their data and request account deletion (Egyptian data protection law 151/2020); money records are kept for legal periods and anonymised	Should
FR-AUTH-11	"Open as" (the CEO or an admin opening a partner's, seller's or factory's view to help) is logged with who, when and why	Must

3.4 Applications and approvals

ID	Requirement	Priority
FR-APP-01	Pitch owner application: owner name, phone, email, national ID number and photos, venue name, governorate, area, address, map pin, number and format of pitches, photos, payout method (bank or InstaPay), acceptance of the partner agreement	Must
FR-APP-02	Seller application: name, store name, type (print on demand / own products / both), national ID number and photos, phone, payout method, about the store, acceptance of the seller agreement	Must
FR-APP-03	Factory application: company name, owner, address, capacity per day, techniques (DTF), contact, bank details, documents	Must

ID	Requirement	Priority
FR-APP-04	Applications go to the **Approvals** queue in the CEO system and in the admin console (for admins with `approvals.*`)	Must
FR-APP-05	Approve opens the dashboard for the applicant and notifies them (in-app + email); reject needs a reason and the applicant can fix and resend	Must
FR-APP-06	New pitches added by an approved owner are "waiting for review" and hidden from the site until approved	Must
FR-APP-07	"Approve new sellers" can be switched off in settings (sellers then go live immediately)	Should
FR-APP-08	ID documents are stored privately (not publicly reachable), viewable only by roles with `kyc.read`	Must

4. Functional requirements — public site and pitches

4.1 Pitches

ID	Requirement	Priority
FR-PITCH-01	A pitch has: name, venue, governorate, area, address, map location, format (5/7/11-a-side), surface, indoor/outdoor, photos, facilities, opening hours per weekday, price per hour (with peak/off-peak), cancellation policy, weekly-package options, status	Must
FR-PITCH-02	Search and filter by area, distance from the user (when location is allowed), format, price, rating, facilities, free hours today	Must
FR-PITCH-03	Pitch page shows live availability for the next 14 days, hour by hour, including offers and the waitlist	Must
FR-PITCH-04	Owners add, edit, close/reopen and delete pitches; deleting a pitch with upcoming bookings cancels and refunds them in full	Must
FR-PITCH-05	Owners block single hours or date ranges (maintenance, private events) with a reason	Must
FR-PITCH-06	Reviews: only customers with a played booking can review; owners reply publicly; owners can report a review; admins hide/restore	Must

4.2 Bookings and checkout

ID	Requirement	Priority
FR-BOOK-01	A customer books one or more consecutive hours; the hour is **locked for 10 minutes** while paying so two people cannot pay for the same hour	Must
FR-BOOK-02	Price = pitch price (with offer) − discount code; a **booking service fee** (3%, editable) is added; all prices **include VAT 14%** (editable)	Must
FR-BOOK-03	Payment: online through the payment gateway, or cash at the venue when the owner allows it	Must
FR-BOOK-04	The injury and conduct waiver must be accepted; the acceptance (version, time) is saved with the booking	Must
FR-BOOK-05	Every booking has a **signed QR ticket** and a PDF receipt; the customer gets an email and an in-app notification	Must
FR-BOOK-06	Cancellation follows the policy saved on the booking (table below); the venue cancelling always refunds in full	Must
FR-BOOK-07	Walk-in bookings added by the owner carry no commission	Must
FR-BOOK-08	Commission on app bookings is 10% (editable per partner by an admin deal); the owner earns the rest	Must
FR-BOOK-09	Online booking money becomes **available** to the owner after the match; available money is settled every **Monday** (editable)	Must
FR-BOOK-10	Waitlist: a customer can wait for a booked hour; when it frees up, everyone waiting is notified with a direct checkout link	Must

ID	Requirement	Priority
FR-BOOK-11	Maintenance mode (set by an admin) pauses new bookings with a message	Must

Cancellation policies (chosen per pitch, saved on each booking):

Policy	Full refund	50% refund	No refund
Flexible	up to 2 h before	until kick-off	—
Standard	up to 6 h before	until kick-off	—
Moderate	up to 24 h before	up to 6 h before	after that
Strict	up to 48 h before	up to 24 h before	after that

The owner keeps the non-refunded share of a late cancellation, minus commission.

4.3 Weekly packages and QR (one QR for the whole package)

ID	Requirement	Priority
FR-PKG-01	The pitch owner chooses which package sizes the pitch offers, from **1 to 12 matches** (for example 4, 8 and 12), and a **discount for each size**	Must
FR-PKG-02	Each package discount is sent to the admin for **approval**; it is not shown to customers until approved. The CEO sets a **maximum package discount** (company policy, e.g. 20%); an admin cannot approve above it	Must
FR-PKG-03	Editing an approved discount sends it back for review; packages already sold keep their original price	Must
FR-PKG-04	The customer picks a size, a weekday and an hour; weeks that are already taken are shown and skipped **before** paying; the whole package is prepaid	Must
FR-PKG-05	The package has **one QR code** for all its matches, with a counter: played / remaining	Must
FR-PKG-06	Each valid scan uses **one** match. A scan is valid only at the right pitch, within the booked hour's window (from 30 minutes before to the end of the hour), and once per scheduled match	Must
FR-PKG-07	A missed match counts as used. An admin can grant **one exception per package** (after agreeing with the pitch manager); a new package allows a new exception	Must
FR-PKG-08	When all matches are used or the last date has passed, the QR expires	Must
FR-PKG-09	The admin and the owner see every scan: time, staff member, device, result (accepted or the reason it was rejected)	Must
FR-PKG-10	Venue memberships (recurring 4-game blocks that renew automatically) are renewed by a nightly job	Should

4.4 QR check-in (single bookings and packages)

ID	Requirement	Priority
FR-QR-01	The QR holds a **signed token** (booking or package id + HMAC signature), never raw data that can be edited	Must
FR-QR-02	Staff scan with the camera, a photo, a USB/Bluetooth scanner, or by typing the booking code	Must
FR-QR-03	The server rejects with a clear reason: edited code, other venue, wrong day or hour, cancelled, already used, package finished, account blocked	Must
FR-QR-04	A successful scan marks the booking "checked in" and the customer's ticket shows it live	Must
FR-QR-05	Scans work on a slow connection: the request is small and idempotent (scanning twice does not use two matches)	Must

4.5 Find players, free slots, ratings and alerts

ID	Requirement	Priority
FR-GAME-01	A customer posts a game (pitch booking, date, players needed, positions, level, share price) and chooses **auto-accept** or **decide manually**	Must
FR-GAME-02	Players ask to join with a position and a message; their share is **paid online and held by Super App** until accepted; declined, withdrawn and removed players are refunded in full	Must
FR-GAME-03	The host sees each player's record (games, attendance, rating, no-shows); accepts, declines, removes, closes requests; a waitlist fills freed spots	Must
FR-GAME-04	After kick-off the host rates players (stars, showed up / no-show); reliability is shown on player cards; admins can block unreliable players	Must
FR-GAME-05	Owners post **free slots** (with an optional last-minute discount) as live cards in "Pitches & free slots"; the card shows Booked/Gone automatically; daily limit per owner (admin setting)	Must
FR-GAME-06	Alerts: a customer follows a pitch or an area with a time window and is notified when a matching free slot is posted	Must

4.6 Tournaments and leagues

ID	Requirement	Priority
FR-TOUR-01	Venues create knockout cups (4/8/16 teams) or leagues (3–10 teams, round robin) with entry fee, prize, dates and hours; match hours are reserved on the pitch automatically	Must
FR-TOUR-02	Teams register and pay online; the waiver is accepted per team	Must
FR-TOUR-03	The organiser draws fixtures and enters results (penalties for cup draws); tables, next rounds and the champion update live	Must
FR-TOUR-04	Cancelling a tournament refunds every team; admins can oversee and cancel any tournament	Must

5. Functional requirements — shop, print on demand and factory

5.1 Shop, stores and orders

ID	Requirement	Priority
FR-SHOP-01	Products belong to a **store**: the official Super App Store, a pitch owner's store, or a seller's own-products store	Must
FR-SHOP-02	A product has name, brand, category, description, specs, photos, sizes, colours, price, compare-at price, stock per variant, low-stock level, status	Must
FR-SHOP-03	Categories are managed by admins (rename, icon, order, hide, add); hiding a category hides its products	Must
FR-SHOP-04	Cart, wishlist, promo codes (store codes and Super App codes), delivery fee (standard 60 / express 120 EGP, free over 1,500 EGP — editable)	Must
FR-SHOP-05	Checkout: address, phone, payment (online or cash on delivery); COD is blocked for customers who must prepay (section 5.2)	Must
FR-SHOP-06	Stock is reserved at checkout and released if payment fails or the order is cancelled	Must
FR-SHOP-07	Store orders move New → Packed → Shipped (courier + tracking) → Delivered, or Cancelled (stock back, refund)	Must
FR-SHOP-08	Store commission: 12% default, lower by seller level (11/10/8%), or a per-partner deal; the official store pays none	Must
FR-SHOP-09	Store money becomes available when the order is delivered	Must
FR-SHOP-10	An admin can hide one product with a reason; the owner is notified	Must
FR-SHOP-11	Product pages show "Sold by" with a link to the store page `/s/{handle}`	Must

5.2 Print on demand (POD)

ID	Requirement	Priority
FR-POD-01	**Catalogue** (managed by admins, made by the factory): products (classic tee, oversized tee, hoodie), base price (includes Super App's margin), factory cost (never shown to sellers), back-print extra, big-size extra (2XL/3XL), colours, sizes, print areas, **details** (fabric, fit, print method, care) and **size chart** (cm)	Must
FR-POD-02	**Design studio:** a seller uploads a **front** and/or **back** design (PNG/JPG/WebP/SVG, ≤ 8 MB), places it on a 3D garment, checks print quality (dpi), picks colours, writes the listing and sets the price	Must
FR-POD-03	Original files are stored in private object storage; the site uses resized previews	Must
FR-POD-04	**Base price** for a design = catalogue base + back-print extra (if the back is printed) − the seller's level discount. The seller's price may **equal** the base (0 profit is allowed) and may not exceed the maximum	Must

ID	Requirement	Priority
FR-POD-05	Adding the back print raises the suggested price by the back-print extra so the profit stays the same	Must
FR-POD-06	Every design goes through **review** (Pro first; Elite publishes instantly with a later spot check); reject needs a reason; editing a live design sends it back to review	Must
FR-POD-07	Customers can report a design (copyright, trademark, offensive); admins take it down or keep it; **copyright strikes** freeze the level for 90 days; 3 strikes suspend the seller	Must
FR-POD-08	Active design limits per level (25 / 100 / 300 / unlimited); bulk upload for Pro and Elite	Must
FR-POD-09	Samples: sellers order their own design; price = base (Starter, Rising), −30% (Pro), one free a month (Elite); paid from the seller's available balance	Must
FR-POD-10	Customers who refuse delivery **2 times** (editable) must pay online for printed items; admins can force on/off per customer	Must
FR-POD-11	The owner of a store sees designs that are not public yet (in review, draft, not approved, paused) on their own store page	Must
FR-POD-12	**Catalogue changes requested by the factory** (price, colours, product pause) are **change requests**: an admin approves or rejects; only approved changes reach seller pages, and sellers using that product are notified	Must

Seller levels (all values editable by admins):

	Starter	Rising	Pro	Elite
Delivered items sold (lifetime)	0	10	50	200
Minimum rating	—	4.0	4.3	4.6
Active designs	25	100	300	unlimited
Base-price discount	0	10 EGP	20 EGP	35 EGP
Store commission	12%	11%	10%	8%
Active discount codes	2	5	15	unlimited
Review	normal	normal	priority	instant + spot check
Samples	base	base	−30%	1 free / month
Fantasy Pass	full price	full price	−50%	free

Level-ups are immediate. A drop below a level's requirements gives a **grace month** at the monthly review; the level goes down only if it is still not fixed at the next review.

5.3 Factory system

ID	Requirement	Priority
FR-FAC-01	Every printed item becomes a **job** routed to a factory, with a deadline (SLA, 48 h default)	Must

ID	Requirement	Priority
FR-FAC-02	Stages: new → printing → quality check → packing → ready. A failed quality check sends it back to printing (reprint at the factory's cost, counted in the defect rate)	Must
FR-FAC-03	Staff roles: owner (everything), supervisor (production, shipping, reports; no money/team), printer, quality check, packer. Every action is logged with the person's name	Must
FR-FAC-04	Problems (out of stock, machine down) alert the supervisor and Super App	Must
FR-FAC-05	When **all items of an order are packed**, a shipment is created with the courier automatically: airway bill, COD amount, pickup request, label PDF with the Super App logo	Must
FR-FAC-06	Courier updates (webhooks) move items to shipped, out for delivery, delivered or refused; customer and seller are notified	Must
FR-FAC-07	Factory money: earnings per piece, deductions, payouts, statement; Excel exports with a dashboard sheet	Must
FR-FAC-08	Settings: capacity per day, SLA, pause intake (printed products are hidden from the shop), pause a product or colour, request a price change (FR-POD-12)	Must
FR-FAC-09	Admins see all factories: late jobs, problems, reassign, cancel, send for reprint after a complaint (with an optional deduction), pay, suspend	Must

5.4 Shipping (courier)

ID	Requirement	Priority
FR-SHIP-01	Couriers sit behind one interface (`IShippingProvider`); the active courier and test/live mode are chosen in the CEO system / admin console	Must
FR-SHIP-02	Create shipment, request pickup, print label, track, cancel, reverse pickup (returns), COD remittance report	Must
FR-SHIP-03	Used by: factory (printed orders), sellers and pitch owners (own-store orders), admins (returns, problems), finance (COD reconciliation), customers (tracking page)	Must
FR-SHIP-04	Until a contract exists, a **simulated courier** produces realistic status updates for testing	Must

5.5 Returns and exchanges

ID	Requirement	Priority
FR-RET-01	Customers request a return within **14 days** of delivery from the order page	Must
FR-RET-02	Store products: return or exchange (another size/colour); "changed my mind" pays the return delivery	Must
FR-RET-03	Printed designs: reprint or refund **only** for a defect or a wrong print (made to order)	Must
FR-RET-04	The seller / store owner (or Super App for the official store and printed items) accepts or rejects within 2 working days; then a courier pickup is booked	Must
FR-RET-05	On receipt: refund to the wallet with a receipt PDF, or ship the replacement; the seller is charged its share; Super App returns its commission; stock goes back	Must

6. Functional requirements — money

6.1 Payments

ID	Requirement	Priority
FR-PAY-01	All online payments go through `IPaymentGateway` (hosted checkout or iframe — **card data never touches our server**)	Must
FR-PAY-02	A payment is confirmed **only** by the gateway's signed webhook (HMAC verified), never by the browser redirect alone	Must
FR-PAY-03	Every payment, refund, top-up, voucher and payout produces a **PDF receipt** with VAT shown inside the amount	Must
FR-PAY-04	Refunds (full or partial) go back through the gateway, or to the customer's wallet when the customer chooses it	Must
FR-PAY-05	Until a gateway contract exists, a **simulated gateway** with test cards is used; the CEO switches the live provider on	Must
FR-PAY-06	Idempotency: retrying a payment or a webhook never charges or credits twice	Must

6.2 Wallets, holds and payouts

ID	Requirement	Priority
FR-WAL-01	Every partner, seller and factory has a wallet with pockets: **pending**, **available**, **settled**. Pitch owners have separate booking and store pockets	Must
FR-WAL-02	Money moves only through a **ledger** (double-entry style rows); balances are computed from the ledger and never edited directly	Must
FR-WAL-03	Holds: booking money after the match; store money on delivery; POD profit **7 days after delivery** (editable)	Must
FR-WAL-04	Withdrawals: minimum **500 EGP**; ID verified before the first one; payout method on file	Must
FR-WAL-05	Withdrawal flow: requested → **prepared** (payables agent or admin) → **sent** (CFO or an admin with `payouts.send`) with a transfer reference → PDF + notification; rejected requires a reason and returns the money to available	Must
FR-WAL-06	**CEO limits** (editable): payouts and settlements over **10,000 EGP** and customer refunds or credits over **1,000 EGP** need the CEO; anything over **50,000 EGP** needs **two approvals** (CEO + CFO). The person who prepared a request cannot approve it	Must
FR-WAL-07	Adjustments (credit or debit a balance) need a reason and are logged; the counterparty is notified	Must
FR-WAL-08	Payouts through the gateway's disbursement API when available; manual bank transfer with a reference otherwise	Should

ID	Requirement	Priority
FR-WAL-09	**Customer wallet top-up:** customers add money to their Super App wallet (quick amounts 200 / 500 / 1,000 or any amount within the limits in settings, default 50 – 10,000 EGP) through the payment gateway; the wallet is credited only after the gateway's signed webhook; a top-up receipt PDF is issued; the wallet balance can pay bookings, orders, game shares, tournaments and the Fantasy Pass; wallet money is not withdrawable as cash	Must

6.3 Points, vouchers and codes

ID	Requirement	Priority
FR-PTS-01	Customers earn points on played bookings, delivered orders and tournament fees (+20 for a review); tiers Bronze / Silver (×1.25) / Gold (×1.5)	Must
FR-PTS-02	**20 points = 1 EGP**; the smallest voucher is **100 EGP = 2,000 points**; points never become cash	Must
FR-PTS-03	Vouchers work on bookings, the shop, printed designs, tournaments, Fantasy Pass and game shares	Must
FR-PTS-04	Admins add or remove points with a reason	Must
FR-CODE-01	**Owner codes** (pitch owners and stores) are funded by the owner and only discount that owner's pitches or products	Must
FR-CODE-02	**Super App codes** (e.g. KICKOFF10, PITCH50, FREESHIP) are funded by Super App; on bookings the partner still earns on the full price	Must
FR-CODE-03	Code options: percent or fixed, minimum spend, total uses, uses per customer, dates, pause/resume; usage, discount given and revenue are reported	Must

6.4 Fees and taxes (all editable in settings)

Item	Default
Booking commission	10% (per-partner deal possible)
Booking service fee (customer)	3%
Store commission	12% (by seller level: 11 / 10 / 8%)
VAT	14%, included in prices
Delivery	standard 60 EGP, express 120 EGP, free over 1,500 EGP
POD hold	7 days after delivery
Minimum withdrawal	500 EGP
Settlement day	Monday
COD refusal limit	2 refusals → prepay

7. Functional requirements — fantasy, communities, support

7.1 Fantasy (Egyptian Premier League and English Premier League)

ID	Requirement	Priority
FR-FAN-01	Clubs, players, positions, fixtures, gameweeks and per-player match stats come from **API-FOOTBALL**; the site never calls it directly — the server syncs on a schedule and caches	Must
FR-FAN-02	**Real player names**; **no club logos** for now (kit colours instead), behind a switch to turn logos on after licensing	Must
FR-FAN-03	Squad of 15 within a budget, max 3 per club, captain/vice, bench, transfers with free transfers and point hits, chips	Must
FR-FAN-04	**Scoring runs automatically after each gameweek** (background job): minutes, goals by position, assists, clean sheets, saves, cards, own goals, penalties, bonus; captain ×2	Must
FR-FAN-05	Points, ranks and group tables update after scoring; late corrections from the data provider are re-applied	Must
FR-FAN-06	Groups: official leagues and private groups with invite codes	Must
FR-FAN-07	**Paid groups** (monthly subscription) stay **hidden** until a payment gateway is live; then admins can create them	Must
FR-FAN-08	Fantasy Pass: 50 EGP/month per league, 80 for both (editable); seller-level discounts	Must
FR-FAN-09	**No cash prizes and no prize pool from entry fees** (to stay clear of gambling law); prizes are sponsor rewards and vouchers	Must
FR-FAN-10	The CEO system shows the API-FOOTBALL **daily usage bar** (requests used / 100 on the free plan): yellow at 70%, red at 90%, with the message "upgrade the plan"	Must
FR-FAN-11	Sponsors, league prizes, announcing winners (voucher codes created automatically)	Must

7.2 Matches and news

ID	Requirement	Priority
FR-MATCH-01	Fixtures, live scores and results (Egyptian league, CAF, Egypt, Premier League) from API-FOOTBALL with caching	Must
FR-NEWS-01	Super App news posts written by admins (headline, summary, body, photo, category, breaking); feed stories can be hidden	Must

7.3 Communities (forum, no private messages)

ID	Requirement	Priority
FR-COM-01	Only Super App creates communities; audience: everyone / sellers / Pro & Elite sellers	Must

ID	Requirement	Priority
FR-COM-02	Posts (text, photo, poll), comments with replies, reactions, pinned posts and announcements, badges (seller level, Super App team, store owner)	Must
FR-COM-03	Super App links become rich cards (pitch, free slot, game, product, design, store, tournament, post); outside links, phone numbers and banned words are blocked	Must
FR-COM-04	Report a post or comment; admins remove, keep, or remove and mute the author; slow mode; muted members can read but not post	Must
FR-COM-05	Admins post and reply as **"Super App Admin"**	Must
FR-COM-06	**No private messages between customers**	Must
FR-COM-07	New posts and comments appear live (SignalR)	Should

7.4 Support and live chat

ID	Requirement	Priority
FR-SUP-01	Customers, partners, sellers and factories open **tickets** (topic, order/booking link, message, photos) from the help center, "Get help" on a booking or order, or their dashboard	Must
FR-SUP-02	Each ticket is a **live chat** with the Super App team (SignalR): typing, read status, attachments; offline replies arrive by email	Must
FR-SUP-03	Disputes show who opened it and against whom; refund with a choice of who pays (Super App, partner, seller); every reply notifies both sides	Must
FR-SUP-04	Auto-reply: "we reply within 24 hours"; priority and assignment (to an admin or an AI support agent)	Must
FR-SUP-05	AI support agents answer within their limits (refunds up to 300 EGP for the frontline agent, 1,000 EGP for the support lead); above that, the ticket is escalated to a human	Must
FR-SUP-06	Help center articles in Arabic and English, search, "was this helpful" votes	Must

8. Functional requirements — platform, admin and CEO

8.1 Notifications and email

ID	Requirement	Priority
FR-NOT-01	**Every action notifies everyone it concerns** (customer, owner, seller, factory, admin) — in-app (bell, live via SignalR) and by email for important events	Must
FR-NOT-02	Notification categories (bookings, orders, designs, money, level, store, account, community, support); users mute categories except account and money	Must
FR-NOT-03	All email is sent from **one official company address** through `IEmailSender`; templates in Arabic and English with the Super App logo	Must
FR-NOT-04	Development: emails are saved as `.eml` files in a local folder; production: SMTP of the company mail provider	Must
FR-NOT-05	Email is sent by a background queue with retries; failures are visible to admins	Must
FR-NOT-06	SMS and WhatsApp through `ISmsSender / IWhatsAppSender` when a provider is contracted	Later
FR-NOT-07	Broadcasts by admins (now or scheduled) to a segment	Must

8.2 Site content

ID	Requirement	Priority
FR-CNT-01	Home banners (up to 3 live, dates, order), announcement bar (tone, link, until)	Must
FR-CNT-02	**Pop-up ads** for Super App services with a 3D object; timing (first, every, stays, per visit), on/off, views and clicks per pop-up	Must
FR-CNT-03	Legal documents (terms, privacy, refunds, waiver, seller agreement, partner agreement, cookies) in Arabic and English, versioned; every acceptance is stored with the version; the CEO marks a version "reviewed by the lawyer"	Must
FR-CNT-04	Cookie consent (essential / analytics) stored per visitor	Must
FR-CNT-05	SEO: per-page description, canonical, Open Graph, JSON-LD; `robots.txt` and `sitemap.xml` generated from live data; private systems are `noindex`	Must
FR-CNT-06	Maintenance mode: pause bookings and/or shop checkout with a message	Must

8.3 Admin console (`/admin`)

Every page and action below requires the matching permission; every action is written to the audit log.

Area	What the admin can do
Overview	KPIs, money through the platform, what needs attention, latest activity

Area	What the admin can do
Approvals	Partners, sellers, factories (with `approvals.*`), new pitches, package discounts, catalogue change requests
Money	Withdrawals (prepare/send/reject), adjustments, revenue report, Fantasy Pass sales, finance approvals (CFO)
People	Customers (block, prepay, notes, points, wallet credit), partners (deal, suspend, open dashboard, hide product), sellers (levels, ID checks, strikes, feature, suspend), factories
Orders & bookings	Everything in one place; cancel and refund; printed items before shipping
Support	Tickets, live chat, disputes
POD	Design review, reports, level review, samples, catalogue, print queue
Factories	Production, couriers, prices, money, teams
Marketing	Super App promo codes, broadcasts
Content	Banners, announcement, pop-ups, news, pickup games, shop categories, help, legal
Fantasy	Leagues, sponsors, prizes, winners, paid groups (after payments go live)
Communities	Reports, posts, communities, muted members, banned words, post as Super App Admin
Tournaments	Oversee and cancel with refunds
Super App Store	Products and orders of the official store
Analytics	Platform analytics with Excel export (dashboard sheet)
Settings	Platform settings (fees, holds, levels, loyalty, delivery, VAT) — admin manager only
Audit log	Read-only, filter, export — admin manager only

8.4 CEO Command Center (`/ceo`)

ID	Requirement	Priority
FR-CEO-01	Overview: live KPIs, department health, "needs your decision", smart alerts; morning briefing	Must
FR-CEO-02	**Approvals inbox:** partner/seller/factory applications and everything over the money limits, with the two-person rule	Must
FR-CEO-03	**Admins:** create, edit, suspend admin accounts; assign roles and permissions, including Approvals; money limits	Must
FR-CEO-04	**Roles & permissions matrix:** create roles, tick permissions per area, see who holds each role	Must
FR-CEO-05	**AI agents panel** (section 9): add, edit, pause, delete agents; tasks, permissions, money limits, schedule, model, reports-to; activity, errors and cost	Must
FR-CEO-06	**Org chart:** a pyramid drawn from the data (humans and AI agents), showing who reports to whom; clicking a box opens the person or agent	Must

ID	Requirement	Priority
FR-CEO-07	**Integrations registry:** every external service (payments, courier, email, SMS, API-FOOTBALL, n8n, AI model, storage, e-receipts) with status, mode (test/live), last call, error rate, usage, on/off switch; secrets are shown only as "set / not set" and can be replaced but never read	Must
FR-CEO-08	**API catalogue:** every Super App endpoint (generated from OpenAPI), grouped by system, with purpose, permission, callers (site, dashboards, agents, n8n), calls and errors in the last 24 h; an endpoint group can be **switched off** with a feature flag (code is never deleted at runtime)	Must
FR-CEO-07a	**New CEO screens in the front end:** `/ceo/integrations` (registry, usage bars, secrets status, test call), `/ceo/api-catalogue` (endpoint groups, switch on/off) and `/ceo/webhooks` (n8n subscriptions, delivery log). They are added to the existing CEO system with **the same design** (same layout, components and styles), in Arabic and English, and linked from the CEO navigation and the Every page map	Must
FR-CEO-09	**Kill switches (9):** bookings, shop, printed designs, payouts, new applications, fantasy, community posting, **joining games (free-slot games)**, AI agents (all or one)	Must
FR-CEO-10	**Limits:** payout, refund, two-person thresholds; maximum package discount; company settings (commissions, fees, points, payout day)	Must
FR-CEO-11	Goals (OKRs), department budgets, compliance calendar, what-if simulator	Should
FR-CEO-12	Reports: finance, operations, customers, partners, sellers, marketing, support, agents — date range, Excel (with a dashboard sheet), PDF	Must
FR-CEO-13	"Open as" any partner, seller, factory staff or admin (logged)	Must
FR-CEO-14	Company audit log	Must

8.5 Public data services (from the public API list)

ID	Requirement	Priority
FR-GEO-01	**Address ↔ map pin** in the partner pitch editor and the pitch-owner application (search an address to drop the pin; moving the pin fills the address), plus factory and seller addresses — through `IGeocoder` (Nominatim now, respecting its usage policy: 1 request per second, identifying user agent, attribution, server-side cache)	Must
FR-GEO-02	**"Near me"** search uses the customer's browser position and distances calculated in SQL from venue coordinates and area centres — no outside API call per search	Must
FR-WX-01	**Weather on the pitch page and at checkout** for outdoor pitches: forecast for the chosen hour up to 7 days ahead (Open-Meteo, cached per area every 3 hours)	Should
FR-WX-02	**Rain / heat alert** in the day-before reminder, suggesting an indoor pitch nearby or another hour	Later
FR-WX-03	**Seller marketing calendar** uses real forecasts (Rising and up) instead of monthly averages	Should
FR-WX-04	Open-Meteo **commercial plan** taken before launch (the free API is non-commercial)	Must (before launch)

ID	Requirement	Priority
FR-HOL-01	**Egyptian public holidays** stored in `ops.PublicHolidays`, synced yearly from the holidays provider (Nager.Date after confirming Egypt coverage, or Calendarific); Islamic holidays saved as **estimated** and confirmed or moved by an admin	Should
FR-HOL-02	Holidays feed partner insights (peak-price suggestions, forecast), the seller marketing calendar, the CEO what-if simulator and a "holiday" badge on availability	Should
FR-VAL-01	**Phone and email validation without an outside API:** phones normalised with libphonenumber to `+20…` (mobile required where SMS is used), emails proven by the confirmation link; optional local block list of disposable email domains	Must

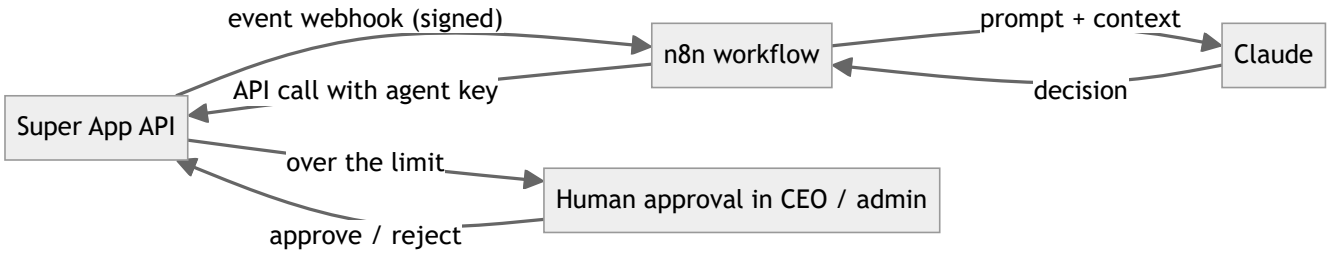
Not used from the same list: betting and odds (illegal in Egypt), currency (EGP only), QR (made on our server), other sports, highlight videos until their terms are checked.

9. AI agents and automation (Claude + n8n)

The company will be run largely by AI agents. The backend must make that **safe, visible and controllable**.

9.1 How agents work with the backend

- Each agent is a **service account** with: name, title, department, reports-to, description, tasks, **permissions** (same model as admins), **money limit**, model (e.g. Claude), schedule, status (active / paused), monthly cost budget.
- Agents call the Super App API with their own **API key** (sent as a header, stored hashed, rotatable, can be revoked instantly). Every call is checked against the agent's permissions and limits exactly like a human admin.
- **n8n** receives **events** from Super App (webhooks) and runs workflows that may call Claude and then call the Super App API back as an agent.
- Anything above an agent's limit becomes an **approval request** for a human; the agent waits for the decision.
- Every agent action is written to the audit log with the agent's name, the input, the result and the cost.



9.2 Requirements

ID	Requirement	Priority
FR-AI-01	CRUD for agents in the CEO system; deleting an agent revokes its keys and keeps its history	Must
FR-AI-02	Tasks per agent: title, description, trigger (event, schedule, manual), expected output, status, last run	Must
FR-AI-03	**Outbox events** for n8n: written in the same database transaction as the change, delivered with retries and an HMAC signature; event types such as `booking.created`, `order.paid`, `order.shipped`, `design.submitted`, `ticket.created`, `payout.requested`, `review.reported`, `fantasy.gameweek.finished`	Must
FR-AI-04	Webhook subscriptions are managed in the CEO system (URL, events, secret, on/off, delivery log)	Must
FR-AI-05	Emergency stop: pause one agent or all agents at once	Must
FR-AI-06	Cost tracking: tokens and money per agent per day; budget alerts	Should
FR-AI-07	An **MCP server** exposing safe Super App tools to Claude	Later

ID	Requirement	Priority
FR-AI-08	**With demo data on** (`SEED__DEMODATA=true`) the seed creates the **18 AI agents** of today's front end (`js/data/org.js`) in their 8 departments, **paused**, with their duties, permissions and money limits, so testing matches the preview. **With demo data off** (real launch) the panel starts empty and the CEO adds the real agents	Must

Separation of duties for agents (from the org design): the payables agent prepares payouts but never sends them; the AI supervisor stays separate so it can watch and stop the others; trust & safety is never merged with customer service.

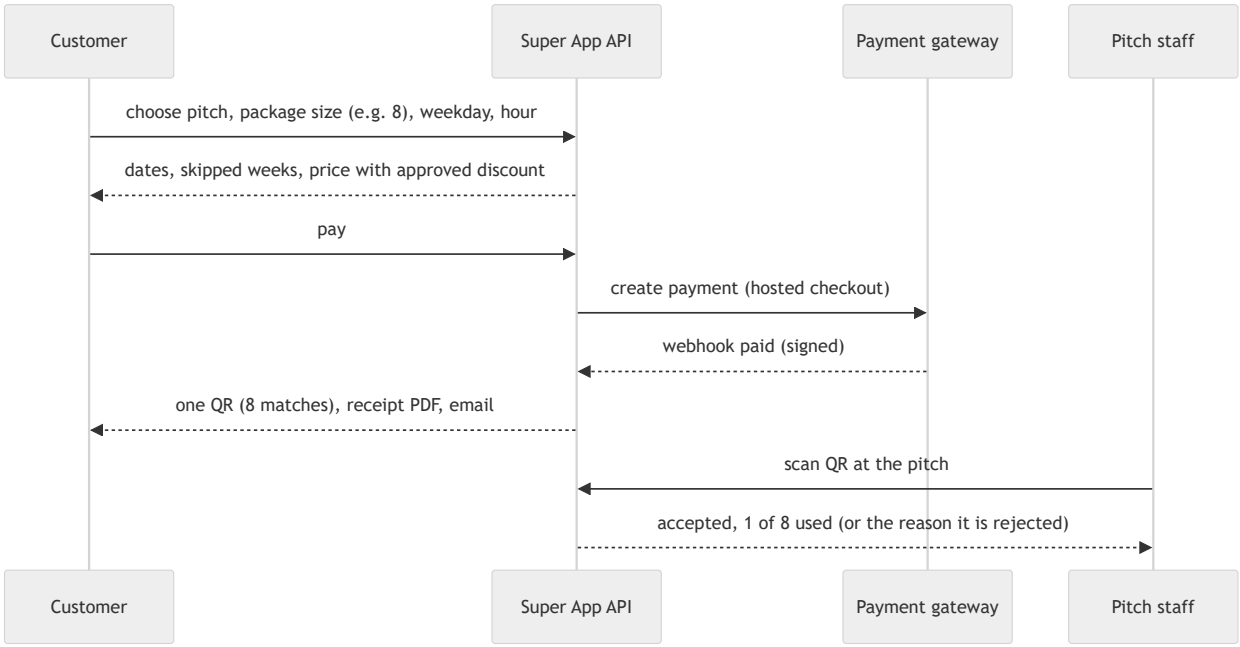
10. Key flows

10.1 A customer becomes a pitch owner

1. The customer opens `/partner/apply`, fills the application (FR-APP-01) and accepts the partner agreement.
2. The application appears in the **CEO Approvals** inbox (and for admins with `approvals.partners`).
3. Approved → the account gets the **Pitch owner** role, the `/dash` dashboard opens, a welcome email is sent.
4. The owner adds pitches, prices, hours, packages and discounts. Each new pitch and each package discount waits for admin approval.
5. Approved pitches appear on the site immediately; bookings flow into the owner's dashboard and wallet.

The same pattern applies to **sellers** (`/sell` → `/seller`) and **factories** (factory application → `/factory`).

10.2 Booking with a package and QR



10.3 A printed-design order, end to end

1. The customer pays (online or COD) → one **POD order line** per item.
2. The line is routed to a factory as a **job** with a deadline; the seller is notified.
3. Factory: printing → quality check → packing. When the whole order is packed, the **courier shipment** is created automatically.
4. Courier webhooks: picked up → out for delivery → delivered (or refused).
5. Delivered → the sale counts for the seller's level; profit is **pending 7 days**, then **available**.
6. Refused → Super App absorbs the cost; the customer's refusal count goes up.

10.4 A payout over the limit

1. A seller requests a 12,000 EGP withdrawal.
2. The payables agent **prepares** it (checks ID, disputes, balance).
3. 12,000 > 10,000 → an approval request goes to the CEO (and, above 50,000, also to the CFO).
4. Approved → sent with a transfer reference → the seller gets a PDF and a notification.

10.5 Fantasy gameweek

1. A scheduled job pulls fixtures and live scores from API-FOOTBALL (cached, inside the free quota).
2. When every match of the gameweek is finished, the scoring job calculates player points, then team points, ranks and group tables.
3. Customers are notified of their points; the `fantasy.gameweek.finished` event goes to n8n.

11. Non-functional requirements

11.1 Security

ID	Requirement
NFR-SEC-01	HTTPS only, HSTS, secure cookies (HttpOnly, Secure, SameSite=Lax/Strict), anti-forgery tokens on state-changing requests
NFR-SEC-02	**Never trust user input:** every request body, route value and query string is validated (FluentValidation) before it reaches a controller action; EF Core parameterised queries only, no string-built SQL
NFR-SEC-03	Authorization on the server for every endpoint: role + permission + **ownership** (a partner can only touch their own pitches, etc.)
NFR-SEC-04	Rate limiting per IP and per account (sign-in, sign-up, password reset, QR scan, chat, uploads)
NFR-SEC-05	Security headers: Content-Security-Policy, X-Content-Type-Options, Referrer-Policy, Permissions-Policy, frame protection
NFR-SEC-06	Uploads: type and size checks, re-encoding of images, private storage for IDs and original artwork, signed time-limited URLs
NFR-SEC-07	Secrets only in environment variables / user-secrets (`.env` in `.gitignore`); integration secrets encrypted at rest; never logged
NFR-SEC-08	Webhooks (payments, courier) verified by signature and replay-protected; outgoing webhooks to n8n signed
NFR-SEC-09	Audit log is append-only; personal data is masked in logs
NFR-SEC-10	Dependency and vulnerability checks before every release; OWASP Top 10 review

11.2 Performance and reliability

ID	Requirement
NFR-PERF-01	API p95 < 300 ms for reads and < 600 ms for writes under normal load; pages stay as fast as today
NFR-PERF-02	Pagination on every list; indexes on every foreign key and every filter/sort column (see ERD)
NFR-PERF-03	Caching for public read data (pitches, products, fixtures, content) with invalidation on change; response compression; static files with long cache headers
NFR-PERF-04	Background jobs (Hangfire) for email, webhooks, fantasy scoring, settlements, holds, renewals, reports — never inside a request
NFR-PERF-05	Health checks (database, storage, background jobs); structured logs (Serilog) with a correlation id per request
NFR-PERF-06	Daily database backups with 14-day retention; restore tested before launch

11.3 Code quality (fixed rules)

- Controllers → Services → Repositories; **dependency injection** for everything; interfaces for services and repositories.
- **Single responsibility:** a method does one thing; more than ~30–40 lines or several unrelated steps → split it.
- **Routes carry no logic**; **middlewares** do authentication, authorization and validation; one **global error handler** turns exceptions into the unified error response.
- **Unified response** for every endpoint:

```
{ "success": true, "data": { }, "message": "Booking created", "errors": null, "traceId": "00-4bf9…"
}
{ "success": false, "data": null, "message": "Validation failed", "errors": { "hour": ["The hour is
already booked."] }, "traceId": "00-9ac2…" }
```

- **Correct HTTP status codes:** 200/201/204 for success, 400 invalid input, 401 not signed in, 403 not allowed, 404 not found, 409 conflict (e.g. hour already booked), 422 business rule broken, 429 too many requests, 500 server error. Never 200 with an error inside.
- Every endpoint is tested in the normal case and the edge cases (Postman collection + automated tests).

11.4 Localisation and accessibility

- Arabic (RTL) and English everywhere; content that customers see is stored in both languages where it is written by Super App (legal, help, catalogue details, notifications, emails).
- Money in EGP, dates in Africa/Cairo time; stored in UTC.
- Keyboard and screen-reader support as today.

11.5 Legal and compliance

- Data protection law 151/2020: consent records, purpose, retention, deletion requests, data-protection licence before launch.
- Consumer protection: 14-day returns for store products; clear prices with VAT.
- No gambling: fantasy has no cash prizes and no prize pool from fees.
- ETA e-receipts: interface ready, switched on after the commercial register and tax card.

12. Data, seed and migration

- **Code-First:** entities and Fluent API configurations define the schema; the owner runs the migrations.
- **Table order:** lookup tables → child tables → junction tables (see ERD).
- **Seed** (idempotent, can run many times): the CEO account (from environment variables), roles and permissions, governorates and areas, shop categories, POD catalogue with details and size charts, seller levels, settings, legal documents, help articles, pop-ups, communities, **and today's demo data** (pitches, products, stores, sellers, designs, orders, bookings, tournaments) so the site is not empty. A switch `Seed:DemoData=false` removes demo data for the real launch.
- The owner then adds real pitches and products through the admin console.

13. Environments

Environment	Where	Database	Email	Payments / courier
Development	Owner's PC (VS Code, `dotnet run`)	SQL Server LocalDB	saved as files	simulated
Testing	Free .NET hosting (e.g. MonsterASP.NET or Azure free tier)	SQL Server (hosted)	placeholder SMTP	simulated / sandbox
Production	**Hostinger VPS** (KVM 2 or larger — shared hosting cannot run .NET or SQL Server) with Docker	SQL Server Express on Linux (10 GB limit, upgrade when needed)	company mail provider	live

14. Delivery plan

Phase	Content
0. Planning	PRD, ERD, API, Integrations, Structure — approved by the owner
1. Foundation	Solution, Domain, DbContext, configurations, seed, Identity, roles/permissions, global error handler, unified responses, logging
2. Accounts & approvals	Sign-up/in, CEO, admins, applications, approvals
3. Pitches & bookings	Pitches, availability, bookings, packages, QR, waitlist, games, tournaments
4. Shop & POD	Stores, products, orders, POD catalogue, designs, studio uploads, factory, courier adapter, returns
5. Money	Payment adapter (simulated), ledger, wallets, holds, payouts, approvals, points, codes, receipts
6. Fantasy & content	API-FOOTBALL sync and scoring, groups, content, pop-ups, communities, support chat
7. Admin & CEO	Every admin page, CEO system, integrations registry, API catalogue, AI panel, org chart
8. Hardening	Security review, performance, backups, tests, Postman, PWA
9. Launch	Real providers, domain and email, data-protection licence, demo data off

15. Risks and open items

Item	Note
Payment gateway	Needs the commercial register and tax card; until then the simulated gateway is used
Courier contract	Simulated courier until a contract; Bosta/Mylerz/Aramex adapters later
API-FOOTBALL free plan	100 requests per day; the seasons it covers must be confirmed on their site before relying on it for the current season
Club logos	Need licensing; hidden behind a switch
SQL Server Express	10 GB per database; plan an upgrade path
AI agents	Costs and mistakes: limits, approvals, emergency stop and logs are mandatory
Official email & domain	Placeholder until the owner provides them

16. Glossary

Term	Meaning
POD	Print on demand: a seller's design printed by the factory when a customer orders
Base price	What Super App keeps for a printed item (factory cost + margin), minus the seller's level discount
Hold	Time before earned money becomes available to withdraw
Package	1–12 prepaid weekly matches at one pitch, one QR
Agent	An AI worker with a service account, permissions and limits
Outbox	A table where events are saved with the change, then delivered to n8n
Two-person rule	Large money actions need two different people to approve